

Maximising Solar PV Energy Penetration

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Crystalline silicon photovoltaic solar modules (Image courtesy: www.tuvdotcom.com)

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Photovoltaics (PVs) will be a key pillar of our future sustainable energy system, and 1:1:1 for wind, solar, and others (hydro, biomass, geothermal) is a reasonable expectation – according to Prof. Eicke R. Weber, director of the Fraunhofer Institute for Solar Energy Systems ISE, and professor for physics/solar energy at the Faculty of Mathematics and Physics and at the Faculty of Engineering at the Albert-Ludwigs-University of Freiburg, Germany.

Crystalline silicon will remain the dominant PV technology. Classical thin-film has to show lower prices or comparable efficiencies. Highly-efficient concentrated photovoltaics (CPVs) will take up a rapidly increasing niche market, competing with con-

centrating solar power (CSP).

Prof. Weber, along with S. Janz and S. Glunz from the Fraunhofer-Institute for Solar Energy Systems ISE and Albert-Ludwigs, University, Freiburg, Germany, presented his thoughts on 'Photovoltaics: Pillar of our Future Sustainable Energy System,' at the recently held *Intersolar Europe 2012* in Munich, Germany.

The technology challenge in PV will be to generate innovations in efficiency and cost reduction fast enough to maintain a profit margin; only players in the XGW-range will survive in the long term. State support for investments in XGW plants, e.g., credit guarantees, might be necessary to maintain globally a level-playing field for PV production.

Creating market barriers will lead to higher prices and less pressure

for cost reduction and innovation, ultimately hurting the economies that adopt them.

Bright future for PV

Globally, the required power is 16 TW today, and it will reach at least 30 TW by 2050. PV is offering at least 10 per cent of this requirement. Optimistically, it should meet 30-40 per cent of the global energy needs. 3TW power corresponds to 12 TW or 12,000 GW of PV capacity.

PV capacity globally installed till 2009 was 20 GW. It was +17 GW in 2010 and increased to +25 GW in 2011. To reach 12,000 GW, we would need almost 500 years at this rate! The PV market will move from a \$50-billion market to hundreds of billion dollars in a few years. This will be accompanied by a drastic cost reduction, making PV one of the most inexpensive ways to produce energy, in the range of 5 cts/kWh, comparable with hydro and onshore wind but less than nuclear and fossil fuels by 2030 or earlier!

For a 100 per cent renewable energy scenario, at least three components or technologies are required in a ratio of 1:1:1. These are solar energy (PV and solar thermal), wind and hydro, and geothermal and biomass.

The maximal sum of PV and wind production was 7.6 TWh in January 2012. The minimal sum was 5.6 TWh in February 2012. The total annual electricity need of Germany is about 600 TWh. The global market forecast is said to be 30 GW in 2014 and 110 GW in 2020. The annual growth rate should be in the range of 20-30 per cent.

If you look at the global PV pro-